My project is a Smart Badminton Training Centre that uses IoT sensors to monitor different things around a badminton facility.

The sensors track court temperature, humidity, lighting, shuttlecock speed, and racket string tension.

I used C# .NET 8, SQL Server, and Entity Framework Core to build the system.

The four parts in my project are the following.

- **Domain:** Contains device classes and checks that the data is valid.
- **Application:** Contains the interfaces and DTOs.
- **Infrastructure:** Connects the system to SQL Server and saves the data.
- **UnitTests:** Tests the code to make sure everything works correctly.
- database has four tables: Devices, TelemetryMetrics, AlertRules, and MaintenanceLogs.

The main goal is to manage badminton sensors, store their readings, check for problems, and keep track of maintenance.

This diagram shows the main classes in my project. DeviceAsset stores the device information, TelemetryMetric stores sensor readings, AlertRule checks for problems, and MaintenanceLog keeps track of repairs.

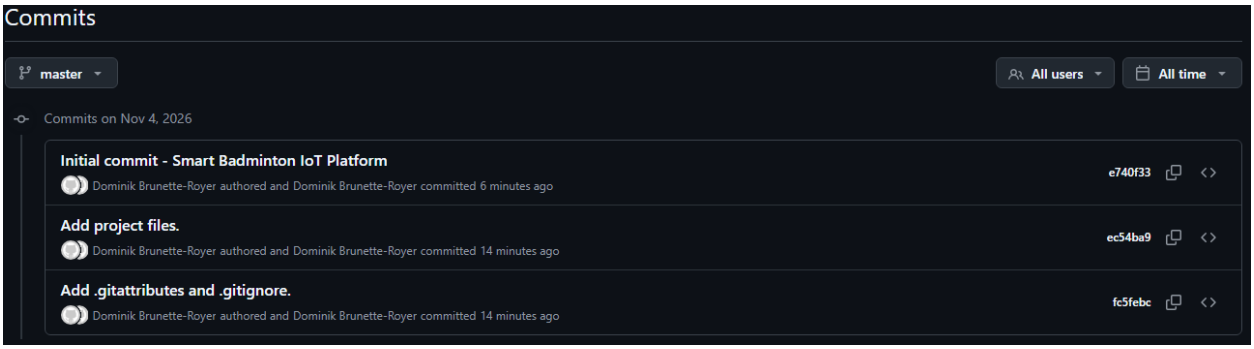
OperatingRange stores the minimum and maximum values allowed for a device.

I completed this project individually.

I worked on the domain classes, application interfaces, SQL Server database, repositories, JSON seed loader, and the unit tests.

I also created the class diagram and ERD for the architecture document.

I used Git to track the changes made to my project.
https://github.com/Dom2329/SmartBadminton-IoTPlatform.git



I created repository interfaces to manage the data in my database.

- **DeviceAssetRepository:** Adds, finds, updates, and deletes devices.
- **TelemetryMetricRepository:** Manages sensor readings.
- **AlertRuleRepository:** Manages alert rules.
- **MaintenanceLogRepository:** Manages maintenance records.

I also created service interfaces for each of these classes.

The interfaces define what the system should do, while the repository classes handle the database operations.

I used Entity Framework Core to connect the repositories to SQL Server.

I used a JSON file called devices.json to add sample devices to my database.

- Court Temperature Sensor
- Shuttlecock Speed Sensor
- Court Humidity Monitor
- Smart Court Lighting Sensor
- Racket String Tension Sensor

I used JsonSeedLoader to read the JSON file and save the devices into SQL Server.

The program also checks for duplicate devices before adding them.

For testing I used MSTest in Visual Studio 2022 to test my project.

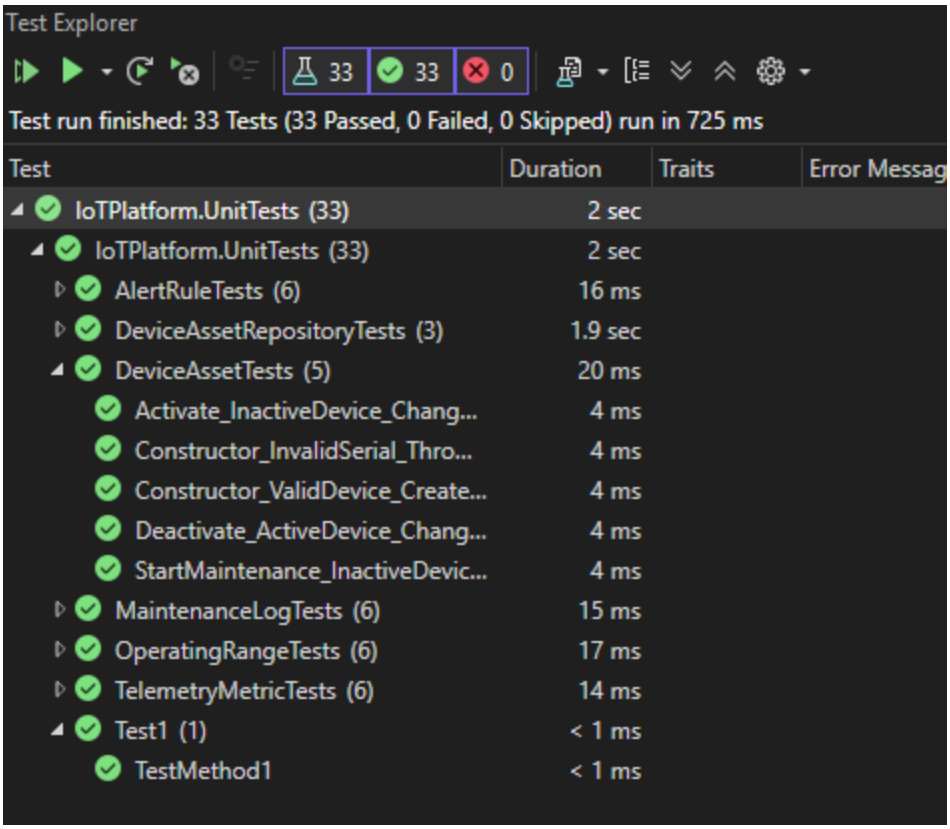
I created tests for each of the following:

- **DeviceAsset:** Checks device information and status changes.
- **OperatingRange:** Checks minimum and maximum values.
- **TelemetryMetric:** Checks sensor readings and invalid data.
- **AlertRule:** Checks if sensor readings trigger alerts.
- **MaintenanceLog:** Checks maintenance information and completion.
- **DeviceAssetRepository:** Tests adding, finding, and deleting devices.

These tests help make sure my classes work correctly and reject invalid data.

I also used an in-memory database to test the repository without changing my SQL Server database.

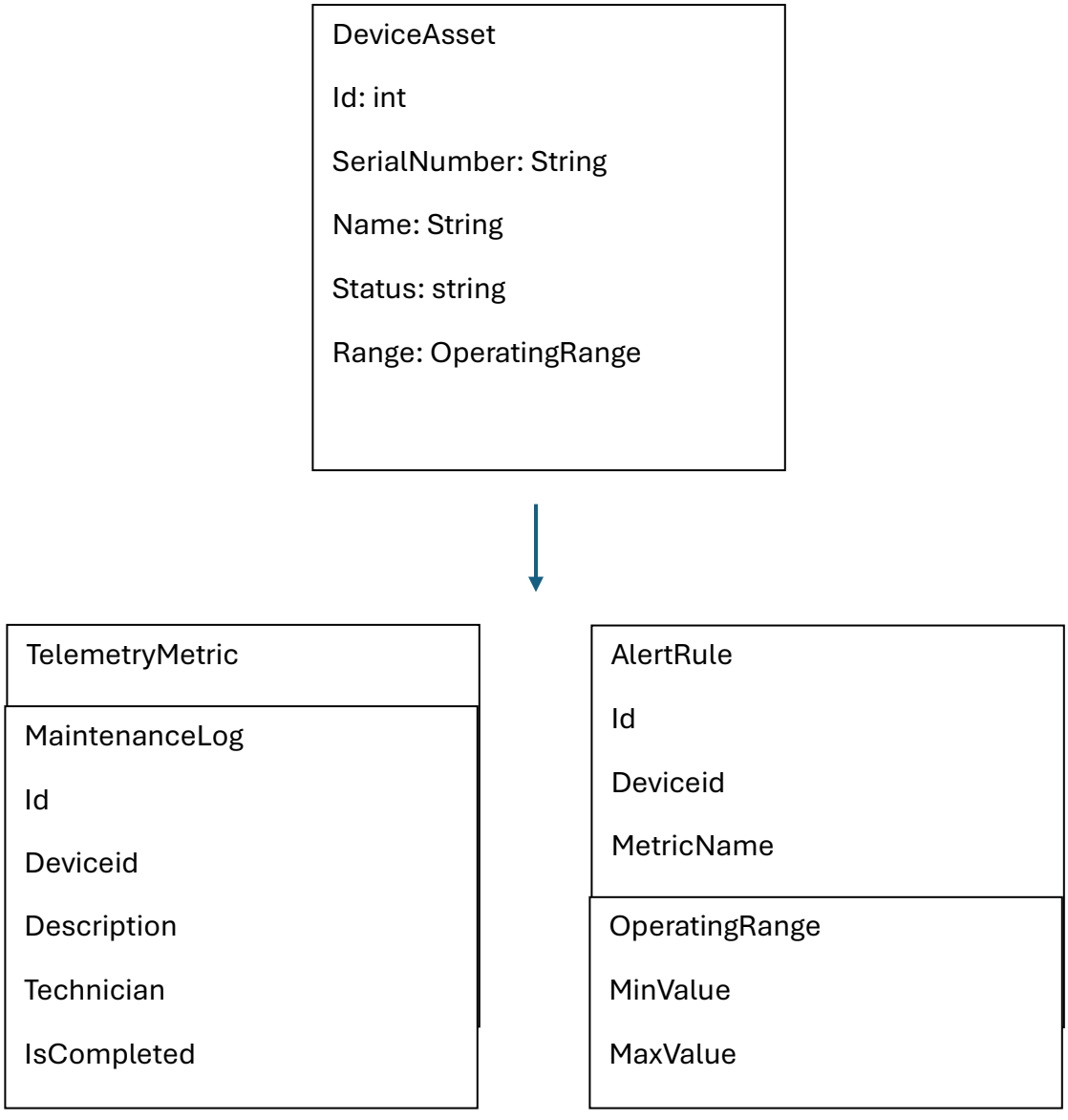
I tested the seed loader and confirmed that the devices were saved successfully.



I would like to improve my Smart Badminton IoT Platform by adding a:

- A dashboard to view sensor readings.
- Live updates from badminton court sensors.
- Automatic alerts when sensor values are too high or too low.
- A maintenance schedule for devices.
- More unit tests to make the system more reliable.

These improvements would make the system easier to use and help monitor the badminton facility.


DeviceAsset
Id: int
SerialNumber: String
Name: String
Status: string
Range: OperatingRange
TelemetryMetric
MaintenanceLog
Id
Deviceid
Description
Technician
IsCompleted
AlertRule
Id
Deviceid
MetricName
OperatingRange
MinValue
MaxValue